Lecture 10

Modular tensor categories and anyon models

Objectives. *Lectures 5 to 7 and 9 built braided fusion categories with a twist: fusion rules, F- and R-symbols, spins. Which of them describe anyons in a two-dimensional medium? Physically, in a bosonic phase every anyon other than the vacuum must be detectable by braiding something around it. Categorically, this is the statement that one matrix, the Hopf link S, is invertible. From S and the spins come the Verlinde formula, which recovers the fusion rules, a projective representation of $\mathrm{SL}(2,\mathbb{Z})$, and a central charge c modulo 8. This lecture computes all of it for Fibonacci and tabulates it for the other models of the course.*

By the end of this lecture you will be able to:

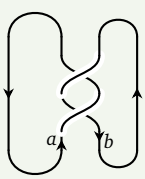
- Compute S and T of a premodular category from its fusion rules, dimensions and spins, and remove a loop encircling a strand.
- Prove that the killing loop projects onto the vacuum when the Müger centre is trivial, and deduce $S^2 = C$ and the criterion for modularity.
- Prove the Verlinde formula and recover fusion rules from S.
- Compute Fibonacci's modular data and central charge, and read c from the Gauss sum for the semion, Ising and the toric code.

10.1 S and T

Definition 10.1 (Premodular category)**.** A *premodular* category is a ribbon fusion category (EGNO Def. 8.13.1): a fusion category with a braiding and a ribbon structure, as in Lecture 9.

As everywhere in the course, the category is unitary with its canonical spherical structure, so $d_a = \mathrm{FPdim}(a) > 0$, $|\theta_a| = 1$, and $\dim \mathcal{C} = \sum_a d_a^2$ as in Lecture 5. A ribbon structure has $\theta_{\bar{a}} = \theta_a$, since $\theta_{a^*} = (\theta_a)^*$ and a scalar is its own transpose.

Definition 10.2 (S- and T-matrices)**.** For simple objects a, b, let $\tilde{S}_{ab}$ be the Hopf link of a and b, oriented oppositely; equivalently of a and $\bar{b}$, both pointing up:

$$\tilde{S}_{ab} \;=\; \text{  } \;=\; \mathrm{Tr}\big(c_{\bar{b},a}\, c_{a,\bar{b}}\big). \tag{10.1}$$

The *S-matrix* is $S_{ab} = (\dim \mathcal{C})^{-1/2}\, \tilde{S}_{ab}$, and the *T-matrix* is $T = \mathrm{diag}(\theta_a)$ (Kitaev eq. (223)).

Proposition 10.3 (S from the data · *proved given coherence*)**.**

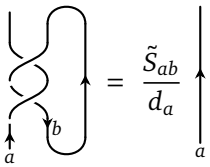
$$\tilde{S}_{ab} = \sum_c N_{a\bar{b}}^c \, d_c \, \frac{\theta_c}{\theta_a \theta_b}.$$

In particular $\tilde{S}_{1a} = \tilde{S}_{a1} = d_a$*,* $\tilde{S}$ *is symmetric, and* S *depends only on* N*,* d *and* θ*.*

Proof. Insert completeness (Equation (5.3)) on $a \otimes \bar{b}$ under the double braiding. On the channel c the double braiding is the scalar $R_c^{\bar{b}a} R_c^{a\bar{b}} = \theta_c/(\theta_a\theta_b)$ (Equation (7.8) and the ribbon relation of Lecture 9). Closing $\psi_c^{a\bar{b}} \psi_c^{a\bar{b}\dagger}$ gives the theta graph $\sqrt{d_a d_b d_c}$ (Equation (5.4)), and the completeness coefficient $\sqrt{d_c/(d_a d_b)}$ turns it into d_c. For the symmetry, dualising $a \otimes \bar{b} \to c$ gives $b \otimes \bar{a} \to \bar{c}$, so $N_{a\bar{b}}^c = N_{b\bar{a}}^{\bar{c}}$, while $d_{\bar{c}} = d_c$ and $\theta_{\bar{c}} = \theta_c$. ∎

Two conventions differ from this one. EGNO, Bakalov–Kirillov, Müger and Rowell–Wang use the unnormalised $\tilde{S}$. EGNO and Müger also take the Hopf link of a and b with the same orientation, and Bonderson's (2.63) sums over N_{ab}^c: either way the result is our $\bar{S}$, the complex conjugate. The two agree for self-dual models, but with $T = \mathrm{diag}(\theta)$ only our S satisfies the relations of § 10.4 in general.

A loop around a strand is a number times the strand, since $\mathrm{End}(a) = \mathbb{C}$; closing the strand shows which number:

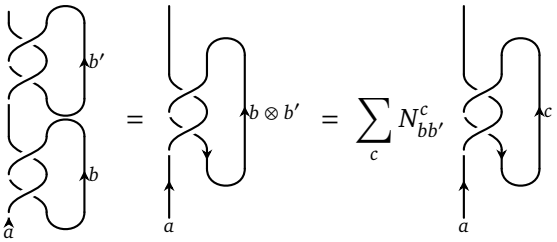


(10.2)

Write $\lambda_a(b) = \tilde{S}_{ab}/d_a$ for this number. Loops can carry combinations of labels, linearly.

Lemma 10.4 (Loops fuse · *proved given coherence*)**.** *For every* a*,* $\lambda_a(1) = 1$ *and* $\lambda_a(b)\,\lambda_a(b') = \sum_c N_{bb'}^c \, \lambda_a(c)$*: the map* $b \mapsto \lambda_a(b)$ *is a character of the fusion ring.*

Proof. Two loops b and b' around a are scalars, so their product is the picture with both. Their fusion is shown with the loop orientations of (10.2):



Slide one loop along the strand a until the two run parallel. By the hexagons and the nested cup and cap of Proposition 4.3 they become one loop labelled $b \otimes b'$. Decompose $b \otimes b'$ by completeness. Each vertex slides around the loop, since the double braiding is natural, until it meets its partner; the bubble $\sqrt{d_b d_{b'}/d_c}$ cancels the coefficient, leaving $\sum_c N_{bb'}^c$ loops labelled c. A loop labelled 1 is empty. ∎

10.2 Premodular, modularity, the killing loop

Lecture 7 called x *transparent* if $c_{y,x}c_{x,y} = \mathrm{id}$ for every y, and the transparent objects the *Müger centre*. A transparent x has $\lambda_x(b) = d_b$ for every b: its loops are those of the vacuum, so no loop around it detects it.

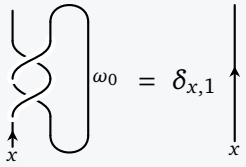
Definition 10.5 (Modular category, anyon model)**.** A premodular category is *modular* if S is invertible (EGNO Def. 8.13.4). A unitary modular category is an *anyon model*: its simple objects are the anyon types, N the fusion rules, and F, R the recoupling and exchange data (Kitaev App. E).

Every gapped phase of bosons in two dimensions, with no symmetry imposed, carries an anyon model, in which only the vacuum is transparent · *physical principle*; this is Kitaev's "braiding nondegeneracy" (pp. 66 and 86). Fermions are excluded: a transparent fermion braids trivially with everything, yet its spin $\theta = -1$ tells it from the vacuum.

The *killing loop* is the loop labelled

$$\omega_0 = \frac{1}{\dim C} \sum_b d_b \, b.$$

Lemma 10.6 (The killing loop · *proved given coherence*)**.** *If the Müger centre of a unitary premodular C is trivial, then for every simple x*



(10.3)

Proof. The loop is the scalar $\Omega_x/\dim C$ with $\Omega_x = \sum_b d_b \lambda_x(b)$. By Lemma 10.4,

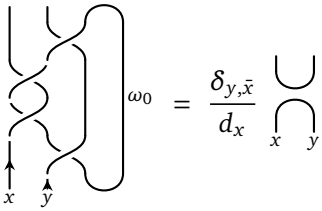
$$\begin{aligned} \Omega_x \lambda_x(b') &= \sum_c \Big(\sum_b d_b N_{bb'}^c \Big) \lambda_x(c) \\ &= d_{b'} \, \Omega_x, \end{aligned}$$

because $N_{bb'}^c = N_{c\bar{b}'}^b$ and $\sum_b N_{c\bar{b}'}^b d_b = d_c d_{b'}$. If $\Omega_x \neq 0$, then $\lambda_x(b') = d_{b'}$ for all b', that is,

$$\begin{aligned} &\sum_c N_{x\bar{b}'}^c d_c \, \frac{\theta_c}{\theta_x \theta_{b'}} \\ &= \sum_c N_{x\bar{b}'}^c d_c. \end{aligned}$$

Each phase has modulus 1 and each $d_c > 0$, so every phase is 1: the double braiding of x with every $\bar{b}'$ is the identity, and x is transparent, hence $x = 1$. Finally $\Omega_1 = \sum_b d_b^2 = \dim C$. ∎

The loop is natural in what it encircles, so around $x \otimes y$ it acts on each channel c of the completeness relation by $\delta_{c,1}$. Only the vacuum channel survives, with coefficient $1/d_x$ and $\psi_1^{x\bar{x}}$ the cup:



(10.4)

The killing loop cuts two strands and reconnects them through the vacuum (Kitaev eqs. (230)–(231); Simon Fig. 17.10).

Theorem 10.7 (Modularity · *cited; proved for unitary C given coherence*)**.** *For a premodular C, S is invertible if and only if the Müger centre is trivial (EGNO Prop. 8.20.12 and Thm. 8.20.7). Then $S^2 = C$, the charge conjugation $C_{ab} = \delta_{\bar{a},b}$.*

Proof. If $x \neq 1$ is transparent, then $\tilde{S}_{xb} = d_x d_b = d_x \tilde{S}_{1b}$: two rows of S are proportional, and S is singular. Conversely, take the trace of (10.4). The right side closes to one loop, $\delta_{y,\bar{x}}$. On the left, the loop b acts on the channel c of $x \otimes y$ by $\lambda_c(b)$, so its trace is

$$\sum_c N_{xy}^c d_c \lambda_c(b) = \sum_c N_{xy}^c \tilde{S}_{bc} = d_b \lambda_b(x) \lambda_b(y),$$

by symmetry and Lemma 10.4. Summing with weights $d_b/\dim C$,

$$\frac{1}{\dim C} \sum_b \tilde{S}_{xb} \tilde{S}_{by} = \delta_{y,\bar{x}}.$$

So $\tilde{S}^2 = \dim C \cdot C$ and $S^2 = C$. ∎

For example, sVec with odd simple s and $\theta_s = -1$ is premodular but every object is transparent (Lecture 7): $S = \frac{1}{\sqrt{2}} \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ is singular. A fermion alone is not an anyon model.

10.3 Verlinde

Theorem 10.8 (Verlinde formula · *cited; first form proved given modularity and coherence*). *In a unitary modular category,*

$$N_{ab}^c = \sum_x \frac{S_{ax} S_{bx} S_{x\bar{c}}}{S_{1x}} = \sum_x \frac{S_{ax} S_{bx} \overline{S_{cx}}}{S_{1x}}.$$

Proof. By the symmetry of S, $S_{ax} = S_{1x} \lambda_x(a)$, with $S_{1x} = d_x/\sqrt{\dim C} > 0$. Lemma 10.4 becomes

$$\frac{S_{ax} S_{bx}}{S_{1x}} = \sum_e N_{ab}^e S_{ex}.$$

Multiply by $(S^{-1})_{xc}$ and sum over x. By Theorem 10.7, $S^{-1} = SC$, so $(S^{-1})_{xc} = S_{x\bar{c}}$, which gives the first form. For the second, $\overline{S_{cx}} = S_{c\bar{x}} = S_{x\bar{c}}$: reversing every crossing of the Hopf link and the orientation of one component is an isotopy, and in a unitary category the inverse of the double braiding is its adjoint (Kitaev eq. (224)). ∎

With the fusion matrices of Lecture 5, $(N_a)_{cb} = N_{ab}^c$, S diagonalises fusion:

$$N_a = S^{-1} \operatorname{diag}\left(\frac{S_{ax}}{S_{1x}}\right)_x S.$$

Its eigenvalues $S_{ax}/S_{1x} = \lambda_x(a)$ are the characters of Lemma 10.4, one for each simple x. The fusion ring is commutative, and S alone determines it. The eigenvalue at $x = 1$ is $\lambda_1(a) = d_a$, the Frobenius–Perron eigenvalue of Lecture 5.

The proof of the first form is graphical at every step: a loop is a number (10.2), loops fuse (Lemma 10.4), and the killing loop exists (Lemma 10.6), which forces $S^2 = C$. Modularity is used once, to make ω_0 kill every non-vacuum strand.

10.4 $\mathrm{SL}(2,\mathbb{Z})$ and the central charge

Theorem 10.9 (Gauss sums · *cited*). *In a modular category, the Gauss sums* $p_\pm = \sum_a d_a^2 \theta_a^{\pm 1}$ *satisfy* $p_+ p_- = \dim \mathcal{C}$ *(Bakalov–Kirillov Cor. 3.1.11, (3.1.20); EGNO Prop. 8.15.4). In the unitary case* $p_- = \overline{p_+}$*, so* $|p_+| = \sqrt{\dim \mathcal{C}}$*, and*

$$\sum_a d_a^2 \theta_a = \sqrt{\dim \mathcal{C}}\, e^{2\pi i c/8}$$

defines the central charge c *modulo* 8.

Theorem 10.10 (Modular data · *cited; $S^2 = C$ proved for unitary C given coherence*). *In a modular category,* $S^2 = C$ *and*

$$(ST)^3 = e^{2\pi i c/8}\, S^2$$

(Kitaev eqs. (232)–(233); Bakalov–Kirillov (3.1.17)). Since $C^2 = 1$ *and* C *commutes with* T*, the matrices* S *and* T *give a projective representation of*

$$\mathrm{SL}(2,\mathbb{Z}) = \langle s, t \mid (st)^3 = s^2,\ s^4 = 1 \rangle.$$

EGNO writes $T = \mathrm{diag}(\theta^{-1})$ (Thm. 8.16.1), and with it the relation reads $(ST)^3 = p_- S^2$ for its unnormalised S: the same content in another convention. The number c is defined only modulo 8: the modular data see it only through $e^{2\pi i c/8}$. Physically it is the chiral central charge of the edge of the phase · *physical principle* (Kitaev eq. (172)).

On a torus, the ground states of a phase with anyon model $\mathcal{C}$ are labelled by its simple objects, four for the toric code (*Quantum Computing*, Lecture 14). The mapping class group consists of orientation-preserving self-diffeomorphisms up to continuous deformation. For the torus this group is $\mathrm{SL}(2,\mathbb{Z})$, and its generators act on the state space by S and T (Simon §17.3.2). Lecture 14 counts these states.

10.5 Fibonacci live; tables of the others

Example 10.11 (Fibonacci's S). Fibonacci has $d_\tau = \varphi$ and $\theta_\tau = e^{4\pi i/5}$ (Lecture 7), so $\dim \mathcal{C} = 1 + \varphi^2 = 2 + \varphi$. By Proposition 10.3, with $\bar{\tau} = \tau$ and $\tau \otimes \tau = 1 \oplus \tau$,

$$\tilde{S}_{\tau\tau} = \theta_\tau^{-2} + \varphi\, \theta_\tau^{-1} = e^{2\pi i/5} + \varphi\, e^{-4\pi i/5}.$$

With $\cos(2\pi/5) = 1/(2\varphi)$, $\cos(4\pi/5) = -\varphi/2$ and $\sin(2\pi/5) = \varphi \sin(4\pi/5)$, the real part is $\frac{1}{2}(\varphi^{-1} - \varphi^2) = -1$ and the imaginary part is 0. So

$$S = \frac{1}{\sqrt{2+\varphi}} \begin{pmatrix} 1 & \varphi \\ \varphi & -1 \end{pmatrix}.$$

Example 10.12 (Fibonacci's modular data). The determinant is $\det S = -1$, so Fibonacci is modular, as Lecture 7 found: the only transparent simple object is 1. And, since τ is self-dual,

$$S^2 = \frac{1+\varphi^2}{2+\varphi} \cdot 1 = 1 = C.$$

Verlinde (Theorem 10.8) returns the fusion rule:

$$N_{\tau\tau}^{\tau} = \frac{\varphi^3 - \varphi^{-1}}{2+\varphi} = 1,$$
$$N_{\tau\tau}^{1} = \frac{\varphi^2 + 1}{2+\varphi} = 1,$$

using $\varphi^3 = 2\varphi + 1$ and $\varphi^{-1} = \varphi - 1$. Sheet 10 checks $(ST)^3 = e^{2\pi i c/8} S^2$ with the c of the next example.

Example 10.13 (Fibonacci's central charge)**.** The Gauss sum is $1 + \varphi^2 e^{4\pi i/5}$. Its real part is $1 - \varphi^3/2 = \frac{1}{2} - \varphi$, and its imaginary part $\varphi^2 \sin(\pi/5) > 0$; its modulus is $\sqrt{2+\varphi}$, as Theorem 10.9 requires. Dividing by this modulus gives

$$\frac{1/2 - \varphi}{\sqrt{2+\varphi}} = -\sin\frac{\pi}{5} = \cos\frac{7\pi}{10},$$
$$\frac{\varphi^2 \sin(\pi/5)}{\sqrt{2+\varphi}} = \cos\frac{\pi}{5} = \sin\frac{7\pi}{10}.$$

Thus the argument is $7\pi/10 = 2\pi c/8$, so

$$c = \tfrac{14}{5} \pmod 8.$$

The conjugate chirality of Lecture 7 has $c = -\frac{14}{5}$.

The other models of the course are tabulated, with c from the Gauss sum; each row is recomputed by `scripts/check-l10-algebra.py`, and Sheet 10 recomputes the toric code, $\mathrm{SU}(2)_k$ and the $\nu/2$ (Kitaev Table 1 and App. E; Bonderson §§5.1–5.6; Simon Table 17.1).

model	d	θ	c
semion $1, s$	$1, 1$	$1, i$	1
Ising $1, \sigma, \psi$	$1, \sqrt{2}, 1$	$1, e^{i\pi/8}, -1$	$\frac{1}{2}$
toric code $1, e, m, \varepsilon$	$1, 1, 1, 1$	$1, 1, 1, -1$	0
Fibonacci $1, \tau$	$1, \varphi$	$1, e^{4\pi i/5}$	$\frac{14}{5}$

Example 10.14 (The toric code's S)**.** For an abelian model every d is 1 and each fusion has one channel, so $\tilde{S}_{ab}$ is the double braiding of a and $\bar{b}$, a sign here. With $\theta = (1, 1, 1, -1)$ on $(1, e, m, \varepsilon)$ and $e \otimes m = \varepsilon$,

$$S = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

The normalised entry is $S_{em} = -\frac{1}{2}$. The unnormalised entry $\tilde{S}_{em} = -1$ is the double-braiding phase *Quantum Computing* measured by taking e around m.

For $\mathrm{SU}(2)_k$ (Theorem 9.9), with labels $j = 0, \frac{1}{2}, \ldots, \frac{k}{2}$,

$$S_{jj'} = \sqrt{\frac{2}{k+2}} \sin\frac{(2j+1)(2j'+1)\pi}{k+2},$$
$$c = \frac{3k}{k+2}$$

(Bakalov–Kirillov Thm. 3.3.20 and Ex. 3.3.22 · *cited*; EGNO Ex. 8.18.5). At $k = 1$ it is the semion row. At $k = 2$, $c = \frac{3}{2}$.

Remark 10.15 (The sixteen-fold way)**.** The braidings with Ising's fusion rules (Theorem 7.9) have $\theta_\sigma = e^{i\pi\nu/8}$ for odd ν, and all eight are modular with $c = \nu/2$ modulo 8 (Kitaev Table 1 and eq. (58); EGNO Ex. 8.18.4). With the eight abelian theories of even ν they form Kitaev's sixteen-fold way (§9).

Remark 10.16 (Ising is not $\mathrm{SU}(2)_2$)**.** Ising and $\mathrm{SU}(2)_2$ have the same fusion rules but different modular data: $\theta_\sigma = e^{i\pi/8}$ and $c = \frac{1}{2}$ for Ising ($\nu = 1$, $\kappa_\sigma = +1$), $\theta_{1/2} = e^{3\pi i/8}$ and $c = \frac{3}{2}$ for $\mathrm{SU}(2)_2$ ($\nu = 3$, $\kappa = -1$). The fusion rules do not determine the anyon model.

Fibonacci's braiding is universal for quantum computation (Freedman, Larsen and Wang, Commun. Math. Phys. 227 (2002) 605–622), beyond this course.

Summary.

- *Handed on:* $\tilde{S}_{ab}$ is the Hopf link, $\sum_c N_{a\bar{b}}^c d_c\theta_c/(\theta_a\theta_b)$, $S = \tilde{S}/\sqrt{\dim C}$ and $T = \mathrm{diag}(\theta)$ (Definition 10.2 and Proposition 10.3); a loop around a is $\tilde{S}_{ab}/d_a$, and loops fuse (Lemma 10.4).
- *Modularity:* with trivial Müger centre the killing loop projects onto the vacuum (Lemma 10.6), so $S^2 = C$; S is invertible exactly when the Müger centre is trivial (Theorem 10.7).
- *Verlinde:* $N_{ab}^c = \sum_x S_{ax}S_{bx}\overline{S_{cx}}/S_{1x}$, and S diagonalises fusion (Theorem 10.8).
- *Central charge:* $\sum_a d_a^2\theta_a = \sqrt{\dim C}\, e^{2\pi ic/8}$ and $(ST)^3 = e^{2\pi ic/8}S^2$ (Theorems 10.9 and 10.10); Fibonacci has $c = \frac{14}{5}$, Ising $\frac{1}{2}$, the semion 1, the toric code 0.
- *Next:* Lecture 11 studies algebras and module categories, the systems on which a fusion category acts.

References

Etingof, Gelaki, Nikshych and Ostrik, *Tensor Categories*: premodular and modular categories and the S-matrix, Defs. 8.13.1–8.13.4 and (8.46), p. 224, and Prop. 8.13.8, p. 225; S^2 and the Verlinde formula, Prop. 8.14.2 and Cor. 8.14.4, p. 227; Gauss sums, Def. 8.15.1 and Prop. 8.15.4, pp. 229–230; $(ST)^3$ with $T = \mathrm{diag}(\theta^{-1})$, Thm. 8.16.1, p. 231; Ising, Ex. 8.18.4, pp. 234–235; c for $C(\mathfrak{g}, q)$, Ex. 8.18.5, p. 235; non-degeneracy and the centralizer, Def. 8.19.1, p. 237, Def. 8.20.1, p. 239, Thm. 8.20.7, p. 241, and Prop. 8.20.12, p. 243. Bakalov and Kirillov, *Lectures on Tensor Categories and Modular Functors*: $\tilde{s}$, (3.1.1)–(3.1.2), p. 47; modular categories, Def. 3.1.1 and Lemma 3.1.4, p. 48; Gauss sums and $(\tilde{s}t)^3$, Lemma 3.1.5 and Thm. 3.1.7, pp. 49–50; (3.1.15)–(3.1.17), p. 51; the killing property, Cor. 3.1.11, p. 52; Verlinde, Thm. 3.1.14, p. 54; $\mathrm{SU}(2)_k$, Thm. 3.3.20 and Ex. 3.3.22, pp. 67–68. Kitaev, *Anyons in an exactly solved model and beyond*, cond-mat/0506438 (v3): Ising-type theories, Table 1, p. 30, and eq. (58), p. 27; the sixteen-fold way, §9, pp. 42–43; the chiral central charge, eq. (172), p. 66; S and Verlinde, eqs. (223)–(228), pp. 84–85; braiding nondegeneracy, §E.5, Def. E.11 and Thm. E.12, p. 86; the killing loop, eqs. (230)–(231), p. 87; $(ST)^3$, eqs. (232)–(233), p. 88. Simon, *Topological Quantum*: S and T, (17.5)–(17.17), pp. 224–226, and the torus, §17.3.2, p. 226; the table of models, Table 17.1, p. 231; the Ω loop, Figs. 17.7–17.11, pp. 230–232. Müger, *Modular categories* (2012), Def. 5.1, p. 13, and S, T, p. 14. Bonderson, PhD thesis (Caltech, 2007): S and T, (2.63)–(2.64), pp. 27–28 (his (2.63) is our $\bar{S}$), and (A.6)–(A.7), p. 96 (his (A.7) is our S); the models, §§5.1–5.6, pp. 77–84. Rowell and Wang, *Mathematics of topological quantum computing*, arXiv:1705.06206, §2.4.2, pp. 11–12.